IoT Homework – Exercise 2

Angelica Ortiz 11191225, Daniel Nägele 11180610

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1 BLE Localization

1.1 Logic

The assumed main use case for the localization of these dentures is the loss and the prevention of loss of said dentures. It makes no sense to use them to localize patients, as not all patients are wearing dentures in general, dentures are not worn at all times, and the fact that the human body will absorb signals. Especially the 2.4 GHz band used by BLE resonates heavily with water molecules and thus the human body. This will lead to absorption, scattering and reflection, and result in severe skewing of RSSI measurements. This leads us to consider the following three scenarios:

In-Mouth: While the denture is worn, we want to put the device to deep sleep and not transmit any beacon signals. Only a signal telling the system that the denture is worn would be useful, but not necessary.

Out of Mouth, Stationary: When the denture is resting on a nightstand or in a glass, only a few initial beacons are necessary to estimate the location. As long as the dentures remain stationary, no further beacons need to be emitted.

Out of Mouth, Moving: If the denture is not worn but still moving, the beacons need to be emitted continuously and often. Such a case would be if the denture is accidentally swept into laundry or the trash and the loss of the prosthesis needs to be prevented. Geo-fencing could be implemented in the sensing side of the system to also automatically alert caregivers.

1.2 Sensing

To detect our scenarios of interest, we need two additional sensors.

- Temperature Sensor: Used to detect if the denture is inside the mouth by measuring the body's heat (e.g. temperature $t > 35\,^{\circ}\mathrm{C}$)
- Acceleration Sensor: Used to detect motion if the denture is outside the mouth.

1.3 Pseudocode

Listing 1 shows the pseudocode for the improved firmware.

Listing 1: Firmware Pseudocode

```
resting_transmits = 0

setup() {
  init_temp_sensor()
  init_accelerometer(Wake_on_Motion_Interrupt) // send interrupt if motion is detected
}

loop() {
  temperature = read_temperature()
  motion_detected = read_last_motion_state()

  if temperature > 35.0:
```

```
    resting_transmits = 0
    disable_ble_beacons()
    // acts as return statement, no further code is executed
    sleep(5 minutes)

  if motion_detected:
    resting_transmits = 0
    enable_ble_beacons()
    set_ble_beacon_frequency(1 Hz)
    sleep(1 second)

  // we have transmitted our beacon 5 times and will not emit any further beacons
      until movement is detected
  if resting_transmits > 4:
    disable_ble_beacons()
    resting_transmits = 0
    // no need to wake up before motion is detected, only 1 day sleep as a fail-safe
    sleep(1 day)
  else:
    enable_ble_beacons()
    // transmit beacon every minute for five minutes
    set_ble_beacon_frequency(1/60 Hz)
    resting_transmits = resting_transmits + 1
    sleep(1 minute)
}
```

1.4 Hardware

The hardware for these dentures must fit inside a dental prosthesis and consume very little power. Thus, the following components were selected.

Nordic Semiconductor nRF52810: This ultra-low power MCU has integrated BLE support and with $2.482\,\mathrm{mm} \times 2.464\,\mathrm{mm}$ a very small package footprint. Additionally, a temperature is already present inside, eliminating the need for an external temperature sensor. Obviously the temperature readings need to be calibrated, due to the dentures surrounding resin and the chip's heat itself.

STMicroelectronics LIS2DW12: This accelerometer draws under $1\,\mu\mathrm{A}$ in active low-power mode. With its similar small package footprint, interrupt support and an integrated temperature sensor, it is the perfect fit for this application. The additional temperature sensor can be used to refine measurements.

1.5 Energy Savings

We assume, that using the RF module and thus the beacon transmission draw the majority of power and that the MCU's and sensor's power draw is negligible.
Original implementation:

$$\begin{aligned} 1\,\mathrm{d} &= 24 \cdot 60 \cdot 60\,\mathrm{s} = 86\,400\,\mathrm{s} \\ 86\,400\,\mathrm{s} \cdot 0.1\,\mathrm{Hz} &= 8640\,\mathrm{beacons} \end{aligned} \tag{1}$$

For our implementation, we assume the following usage characteristics over a single day:

- In-Mouth: 16 h
- Outside Mouth Moving: 10 min
- Outside Mouth Resting: 7 h50 min

Improved implementation:

$$\begin{aligned} n_{mouth} &= 0\,\text{beacons} \\ n_{resting} &= 5\,\text{beacons} \\ n_{moving} &= 10\,\text{min} \cdot 1\,\text{Hz} = 600\,\text{beacons} \\ n_{total} = n_{mouth} + n_{resting} + n_{moving} &= 605\,\text{beacons} \end{aligned} \tag{2}$$

This results in the following energy savings:

$$605 \div 8640 \approx 0.07 = 7\% \tag{3}$$

Despite our aggressive beacon transmission during movement, we transmit 7% of the original beacon count and thus only use 7% of the original power draw.